Project : Real-Time IoT Device Monitoring & Cloud Dashboarding Platform
Group : Group 10
Students : Nathan Coudray, Thoma Chalon, Basile Gasser
Git URL (if applicable) : https://github.com/chatLong2000/cloud_project_2_grp10
Link to a video : https://www.swisstransfer.com/dl/01a19cf2-0be2-71b8-b045-973622f74457

Architecture

1. A figure detailing the architecture

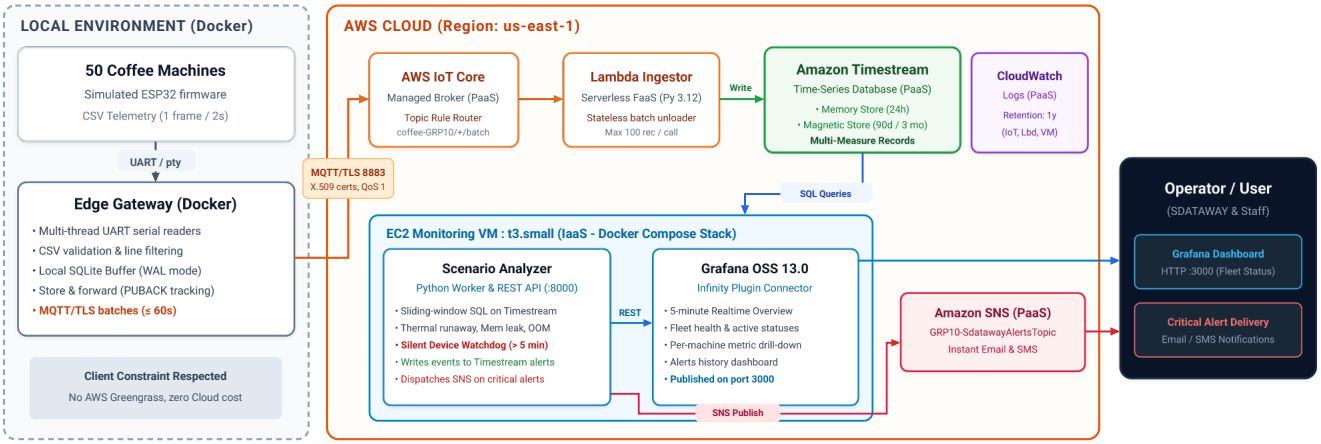


Figure 1: End-to-End IoT Pipeline: Data Ingestion, Multi-Measure Time-Series Storage, Scenario Analysis, and Monitoring

2. For each component: two lines of text (maximum) explaining the component

- **50 Coffee Machines (Simulators):** 50 concurrent Python processes simulating ESP32 microcontrollers, streaming 8-field CSV telemetry frames every 2 seconds over virtual serial UART ports (`socat` PTY tunnels).
- **Edge Gateway (Docker):** Local multi-threaded service reading serial streams, validating CSV metrics, buffering records in SQLite (WAL mode, store-and-forward), and publishing MQTT batches (≤ 60 s).
- **AWS IoT Core (MQTT Broker):** Managed serverless broker securing edge communication via mutual TLS (X.509 certificates) and routing incoming batches on `coffee-GRP10/+/batch` via an SQL Topic Rule.
- **AWS Lambda Ingestor:** Stateless Python FaaS function triggered per batch, packaging measures into Multi-Measure records and batch-writing up to 100 items per API call into Amazon Timestream.
- **Amazon Timestream (Database):** Serverless time-series database (`SdatawayIoTDB`) holding telemetry and alerts, automatically tiering data from Memory Store (24h) to Magnetic Store (90 days / 3 months).
- **Scenario Analyzer Worker:** Continuous Python analytics engine on EC2 running sliding-window SQL queries to detect anomalies (`thermal_runaway`, leaks, OOM) and silent machines ($>$ 5 min).
- **Grafana OSS 13.0 Dashboard:** Web visualization server hosted on EC2 with the Infinity plugin, presenting 5-minute real-time fleet health, active statuses, and per-machine metric drill-downs.
- **Amazon SNS (Alerting):** Managed Pub/Sub notification service (`GRP10-SdatawayAlertsTopic`) instantly fanning out critical alarm notifications to operating staff via email and SMS.
- **Amazon CloudWatch Logs:** Centralized logging service capturing system, broker, Lambda, and worker execution logs with an enforced 1-year (365 days) audit retention policy.

Deployment

For each component, the specific Cloud service and deployment model are detailed below:

- **Edge Gateway & Simulators (Local Docker Environment):** Deployed locally in Docker containers (`docker compose`). Running the Edge outside AWS fulfills SDATAWAY's physical Raspberry Pi constraint, validates offline store-and-forward persistence across network outages, and incurs $0 in cloud runtime costs.
- **AWS IoT Core (PaaS):** Managed serverless MQTT broker in AWS `us-east-1`, secured via client X.509 certificates and an IoT Policy restricting publishing to `coffee-GRP10/+/batch`. Automatically routes batches to Lambda via an SQL Topic Rule.
- **AWS Lambda Ingestor (FaaS):** Stateless event-driven serverless function (`Python 3.12`, x86_64, 128 MB RAM, 15 s timeout). Scales dynamically with zero idle cost; execution IAM role is strictly scoped to `timestream:WriteRecords` and CloudWatch logging.
- **Amazon Timestream (PaaS — Time-Series Database):** Serverless time-series database (`SdatawayIoTDB`) with `telemetry` and `alerts` tables using multi-measure records. Uses automated dual-tier storage: 24-hour Memory Store for real-time dashboards and 90-day Magnetic Store ($0.03/GB/mo) for 3-month retention.
- **Scenario Analyzer & Grafana OSS (IaaS — Amazon EC2 VM):** Deployed on a single Amazon EC2 `t3.small` instance (Amazon Linux 2023) in the default VPC, hosting a containerized Docker Compose stack (`analyser` worker + `grafana` OSS 13.0):
  - *Access & Security:* Managed via AWS Systems Manager (SSM Session Manager) with SSH (port 22) disabled. IAM Instance Profile grants least-privilege Timestream query, SNS publish, and CloudWatch logging rights.
  - *Networking:* Grafana is served on port 3000 with ingress restricted to authorized operator CIDRs via Security Groups. Grafana admin credentials are securely managed through SSM Parameter Store.
- **Amazon SNS & Amazon CloudWatch Logs (PaaS):** SNS topic `GRP10-SdatawayAlertsTopic` dispatches email alerts for critical scenarios; CloudWatch captures IoT Core, Lambda, and Analyzer logs with an enforced 365-day (1-year) audit retention schedule.

Tasks

List of tasks realised within the project, their concrete outcomes, and student involvement:

Tasks	Outcome	Students involved
1. Initial Architecture Definition	Designed the end-to-end cloud pipeline (`ESP32 -> UART -> Edge -> MQTT -> IoT Core -> Lambda -> Timestream -> VM/Grafana/Analyzer -> SNS/CloudWatch`) meeting SDATAWAY constraints and SLAs (< 5 min latency).	All three students
2. Database Technology Selection & Trade-Off	Researched and benchmarked storage options (Amazon Timestream vs. DynamoDB, InfluxDB), establishing the multi-measure schema and 3-month cost/sizing model ($20–$35/month).	Basile Gasser
3. Coffee Machine Fleet Simulation	Containerized 50 concurrent ESP32 simulators generating 8-field CSV telemetry frames (0.5 Hz) over `socat` virtual serial PTY channels.	Nathan Coudray
4. Edge Gateway (Serial, Buffer & MQTT)	Developed multi-threaded Docker gateway reading serial streams, validating CSV metrics, buffering in SQLite (WAL mode, store-and-forward), and publishing MQTT batches (≤ 60 s).	Nathan Coudray
5. AWS IoT Core Ingestion & CloudWatch	Configured managed IoT Core broker, X.509 mutual TLS authentication, topic routing rules (`coffee-GRP10/+/batch`), and broker connection logging in CloudWatch.	Nathan Coudray
6. AWS Lambda Ingestor Component	Built and deployed stateless Python FaaS `ingestor` unpacking incoming MQTT batches and inserting records via the Timestream `WriteRecords` batch API (up to 100 records/call).	Basile Gasser
7. Amazon Timestream Database Deployment	Automated provisioning of database `SdatawayIoTDB` with `telemetry` and `alerts` tables, configuring lifecycle retention (24h Memory Store, 90d Magnetic Store).	Basile Gasser
8. Scenario Analyzer (Detection Engine)	Python worker querying Timestream with stateful per-machine detectors (thermal runaway, leak, OOM, silent > 5 min) and a JSON API for Grafana.	Thoma Chalon
9. Detector Validation	Seeded replay generator with ground-truth fault log, precision/recall scoring per scenario and Jupyter notebooks for threshold tuning; runs without AWS.	Thoma Chalon
10. Grafana Dashboard	Infinity datasource and fleet dashboard provisioned as code (health, alerts, per-machine drill-down); same Compose stack locally and on the VM.	Thoma Chalon
11. Monitoring VM Provisioning (boto3)	Notebook automating the EC2 `t3.small` deployment: instance profile, Security Group (port 3000), SSM, code bundle, redeploy, teardown.	Thoma Chalon
12. Amazon SNS Alerting System	Provisioned topic `GRP10-SdatawayAlertsTopic` with email/SMS subscriptions and integrated the alert dispatching mechanism in the Scenario Analyzer for critical fault notifications.	Basile Gasser
13. CloudWatch Logging for DB & Analyzer	Implemented centralized CloudWatch logging for Timestream ingestion tracking and Scenario Analyzer application logs, enforcing the mandatory 365-day (1-year) retention policy.	Basile Gasser
14. Demonstration Video	Recorded and edited the platform video demonstration showcasing telemetry generation, real-time edge batching, cloud storage, anomaly detection, and Grafana dashboarding.	All three students
15. Project Report (Typst Documentation)	Authored the 2-page project documentation in Typst according to the template, detailing architecture, component roles, cloud deployment models, and task breakdown.	All three students